

YASH DEODHAR

5400 North Lake Road, Merced, California, 95343

☎ 773-828-2016 ✉ yashd1618@gmail.com [in www.linkedin.com/in/yash-deodhar](https://www.linkedin.com/in/yash-deodhar) [github https://github.com/Yash-Deodhar](https://github.com/Yash-Deodhar)

Education

University of California Merced

PhD in Applied Mathematics

Department of Applied Mathematics - 4.0/4.0 GPA

Aug 2025 – Present

Merced, California

Northwestern University

Master of Science in Applied Mathematics

Department of Engineering Sciences and Applied Mathematics - 3.89/4.0 GPA

Sep 2022 – Dec 2023

Evanston, Illinois

Manipal Institute of Technology

Bachelor of Technology with Honors (minor specialization in computational mathematics)

Department of Electronics and Communication Engineering - 8.68/10 CGPA

Aug 2018 – June 2022

Manipal, India

Publications

1. Y. Deodhar, J. J. S. Reddy, P. D. Kakade, and R. Kakade, **Investigation of different transfer functions for optical limiting amplifier used in a 2R burst mode optical regenerator**, International Journal of Electronics and Telecommunication (2021) [Link](#)
2. Y. Deodhar and P. D. Kakade, **BER Evaluation in the cascade of 2R BMORs in the presence of Atmospheric Turbulence**, IEEE Workshop on Recent Advances in Photonics (2022) [Link](#)
3. Jithin D. George, Brianna N. Ruggiero, Yash S. Deodhar, Linsey C. Seitz, Niall M. Mangan, **Modeling and numerical simulation of the electrocatalytic reaction environment for the Oxygen Reduction Reaction** (In Progress)

Research Experience

Investigating trends across ions in Electrocatalytic Oxygen Reduction Reaction

RA with Professor Niall Mangan (Northwestern University)

April 2024 - August 2025

- Derived a novel mathematical model for calculating current due to 1 reaction in a system with 2 competing reactions
- Proved that steric effects are necessary to model ion interactions, ions previously considered to be point charges
- Identified areas for code optimization in simulation code by profiling it, modified functions to implement JIT compiler
- Currently investigating the effect of different ions in electrolyte on the current produced

Analysis of Burst Mode Optical Regenerators (BMORs)

with Thesis Advisor Professor Priyanka Kakade (Manipal Institute of Technology)

Aug 2020 - June 2022

- Conducted a literature review, categorized existing optical limiting amplifier (OLA) designs based on technology used
- Simulated a cascade of BMORs and analyzed the bit error rate to find optimal transfer function for OLA
- Used optimal transfer function found to model a cascade of BMORs to evaluate their performance in a hybrid optical communication system

Projects

- Fast Inverse Square Root Algorithm [Link](#)
- Van Der Pol Oscillator (Serial) [Link](#)
- Complex Grinsburg Landau (Serial, MPI) [Link](#)
- Diffusion Equation (MPI, CUDA) [Link](#)
- Rock Paper Scissors model using Alternating Direction Implicit Method (CUDA) [Link](#)
- Calculating Nth Fibonacci number using fftw library
- Parallel implementation of Ramanujam-Sato formula using gmp library to obtain arbitrary digits of pi
- Fourier Neural Operator for PDE Inverse Problem

Technical Skills

Languages: Python, MATLAB, C, C++, Julia

Leadership Roles

LoopMIT

May 2020 - June 2021

Levitation Subsystem Senior Member

Manipal Institute of Technology

- Used Fusion 360 and AutoCAD to design a testing rig to measure magnetic lift and drag forces
- Investigated flux patterns of different arrangement of magnets using FEM software
- Simulated magnetic lift and drag forces using COMSOL and ANSYS

Chess Club

May 2020 - June 2021

President

Manipal Institute of Technology

- Conducted the first open tournament in Manipal Institute of Technology
- Hosted and streamed the first online intersection tournament for freshmen classes in Manipal Institute of Technology
- grew the club size from ~50 to ~150 members

Miscellaneous Achievements

- Top 500 rank in SOF's International Mathematics Olympiad (95.12 percentile)
- Gold medal for achieving top 100 rank in SOF's International English Olympiad
- Secured first place in a city level General Knowledge competition